

Linux Security Hardening Assessment

System configuration review, risk analysis, recommended remediation, and validation

Scope

Representative Linux security assessment covering accounts, privilege, SSH, permissions, firewall, services, logging, patch posture, scheduled tasks, and file-system risks. Commands and outputs are portfolio lab examples rather than claims about a production employer environment.

Control area	Review command / evidence	Observed lab state	Risk	Recommendation	Validation
Privileged accounts	awk -F: '(\$3==0)'/etc/passwd	Only root has UID 0.	Low	Maintain single authorized UID 0 account; use sudo for administration.	Re-run UID 0 query and review sudo group.
SSH root login	sshd_config	PermitRootLogin no	Low	Keep root SSH login disabled.	sshd -T confirms permitrootlogin no.
SSH password auth	sshd_config	PasswordAuthentication no	Low	Use public-key authentication; protect private keys.	sshd -T + successful key login.
SSH brute-force resistance	MaxAuthTries	MaxAuthTries 4	Medium	Keep low retry count; centralize auth logs; add rate controls where appropriate.	Failed-login test produces expected log/alert.
Firewall	ufw/firewalld rules	Active; SSH limited to admin subnet.	Low	Keep default deny; document business-required ports.	Port scan from unauthorized test segment fails.
World-writable files	find ... -perm -0002	No unexpected files in assessed paths.	Low	Repeat scheduled review; remediate unauthorized writable files.	Re-run scan after remediation.
Services	systemctl enabled services	SSH, cron, rsyslog enabled in sample.	Medium	Disable unused services; assign owner/business reason for required services.	Enabled-service list matches approved baseline.
Logging	rsyslog/journald	Core logging enabled.	Medium	Forward auth/admin/security events to central platform and retain per policy.	Search central log source for test event.
Patch posture	package manager / update status	Representative lab marked current at review.	Medium	Apply risk-based patch SLA and track exceptions.	Package scan reports no overdue critical updates.
Scheduled tasks	crontab/system cron	Only approved maintenance tasks in sample.	Low	Review new/modified cron entries and owners.	Compare to approved task inventory.

Hardening Priorities

- Protect administrative access first: SSH, sudo, privileged accounts, and MFA/strong access controls where supported.
- Reduce attack surface by disabling unnecessary services and restricting inbound network paths.
- Centralize authentication and administrative logs so hardening controls can be monitored.
- Use repeatable validation commands after every corrective action.
- Treat CIS guidance as a benchmark to evaluate applicability, not a script to apply blindly without service-impact review.